EC442: Lecture Notes

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Notes on the 1st year Macro class

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1. Lecture 1 Sep. 29

1.1. Why heterogeneity

Complete markets + standard welfare assumptions:

- Competitive eq. Pareto efficient; planner representation w/ fixed weights $\alpha_i > 0$.
- Common discounting; preferences $u(c) - v(n)$. Consumption FOCs:

$$\alpha_i u'\left(c_t^i\right) = \alpha_j u'\left(c_t^j\right) \Rightarrow \frac{u'\left(c_t^i\right)}{u'\left(c_t^j\right)} = \frac{\alpha_j}{\alpha_i}. \tag{1.1}$$

  Marginal-utility ratio constant across time/states.
- Common CRRA: $u'(c) = c^{-\sigma}$.

$$\frac{c_t^i}{c_t^j} = \left(\frac{\alpha_i}{\alpha_j}\right)^{\frac{1}{\sigma}}. \tag{1.2}$$

  Fixed consumption shares → **perfect insurance** against idiosyncratic income shocks. Aggregate risk remains; unequal Pareto weights → unequal consumption.
- Strictly convex labor disutility; interior labor choice; aggregate conditions fixed: productivity ↑ → labor ↑; consumption share unchanged.

Implications:

- Karl Marx
- Full insurance at odds w/ data

> **Lecture addition: Intuition for Complete Markets**
>
> - **Example**: 18 students; one high earner, others lower. Complete markets → high earner sells state-contingent claims → consumption decoupled from income.
> - **Marx analogy**: "From each according to ability, to each according to needs." Efficient allocation equalizes marginal utilities regardless of income.
> - **Empirical mismatch**: Data shows strong income-consumption correlation. Complete markets imply correlation vanishes → model unrealistic. Heterogeneity requires market incompleteness.
>
> Recording: 00:13:54, 00:16:04

1.2. How to model incomplete markets?

1. **Exogenous**: assume some markets absent.
    1. e.g. save/borrow via noncontingent bonds; no state-contingent claims.
2. **Endogenous**: derive market incompleteness from frictions.
    1. e.g. limited commitment (participation constraints), private information.
    2. deeper but harder to do

Lecture addition: Course Roadmap

- **Methods**: Dynamic programming, numerical methods.
- **Background**: Equilibrium and welfare theorems.
- **Growth**: Basic models of economic growth.
- **Incomplete markets**: Endogenous (participation constraints/limited commitment), exogenous (asset structure), search models.

Recording: 00:31:25, 00:33:02

Lecture addition: Frictions for Endogenous Incompleteness

- **Limited commitment**: Contracts unenforceable. E.g., US car insurance limits; agents flee jurisdiction (e.g., to Frankfurt) to avoid liability → limits risk trading.
- **Private information**: Effort/talent unobservable. Income sharing → moral hazard (effort reduction) → complete insurance inefficient/impossible.

Recording: 00:22:10, 00:27:38

1.3. Planning problem

Set $n_t = A = 1$; $f(k) = F(k, 1)$. Given $k_0 \geq 0$. Discount factor $0 < \beta < 1$; depreciation $0 \leq \delta \leq 1$.

Choose sequences $c_t, k_{t+1} \geq 0$:

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + k_{t+1} = f(k_t) + (1-\delta)k_t. \tag{1.3}$$

- **Lagrangian**; current-value multipliers λ_t:

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t [u(c_t) + \lambda_t (f(k_t) + (1-\delta)k_t - c_t - k_{t+1})]. \tag{1.4}$$

- **Interior FOCs**:

$$\begin{aligned} c_t &: u'(c_t) = \lambda_t, \\ k_{t+1} &: \lambda_t = \beta \lambda_{t+1} [f'(k_{t+1}) + 1 - \delta]. \end{aligned} \tag{1.5}$$

  Euler equation:

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta]. \tag{1.6}$$

- **Transversality condition**:

$$\lim_{T \to \infty} \beta^T \lambda_T k_{T+1} = \lim_{T \to \infty} \beta^T u'(c_T) k_{T+1} = 0. \tag{1.7}$$

  Discounted shadow value of terminal capital → 0.
- Increasing, concave u; concave f; finite discounted utility: feasibility + FOCs + TVC → global optimum. At corners, use KKT conditions.

1.4. Towards a recursive formulation

- Sequence approach: infinitely many dated choices + optimality conditions.
- Repeated trade-off: consume today vs. save for tomorrow.
- Stationary problem; current capital k_t summarizes relevant history.
- Recursion: continuation value summarizes future choices; same decision problem at every date.

Value function: maximal lifetime utility from initial capital k_0:

$$v^*(k_0) = \max_{\{c_t, k_{t+1}\}_{t\geq 0}} \sum_{t=0}^{\infty} \beta^t u(c_t). \tag{1.8}$$

Feasible sequences: $c_t, k_{t+1} \geq 0$; $c_t + k_{t+1} = f(k_t) + (1-\delta)k_t$, all $t \geq 0$.

Split date 0 from continuation:

$$v^*(k_0) = \max_{c_0, k_1} \left[u(c_0) + \beta \max_{\{c_t, k_{t+1}\}_{t\geq 1}} \sum_{t=1}^{\infty} \beta^{t-1} u(c_t) \right]. \tag{1.9}$$

- Outer max: $c_0, k_1 \geq 0$; $c_0 + k_1 = f(k_0) + (1-\delta)k_0$.
- Inner max: given chosen k_1, same feasibility constraints for all $t \geq 1$. Discounting restarts at date 1 → inner value $v^*(k_1)$.

Continuation value absorbs future choices. **Bellman equation** (current capital k, next-period capital k'):

$$v^*(k) = \max_{c, k'} [u(c) + \beta v^*(k')] \tag{1.10}$$

subject to

$$c + k' = f(k) + (1-\delta)k, c, k' \geq 0. \tag{1.11}$$

Infinite sequence → one-period choice + discounted continuation value. Stationarity → same problem at every date.

Bellman equation = **functional equation**; unknown = function $v(k)$.

Questions:
- **Existence**: conditions for solution v?
- **Uniqueness**: conditions for unique v within chosen function class?
- **Equivalence**: when does $v = v^*$, with maximizing policy solving original sequence problem?

Answer:
- *Recursive Macroeconomic Theory* (RMT), "Dynamic Programming" (Ljungqvist & Sargent, 2018, ch. 3).

Lecture addition: Computational Advantage of DP

- **Lagrangian**: Leads to 2nd-order difference eq. Solving via "shooting" → local, clunky.
- **Recursive**: Bellman eq. → direct computation of **global solution** via iteration.
- **Convergence**: VFI guaranteed to converge to unique fixed point (via CMT), unlike many ODE/PDE methods.

Recording: 01:06:37, 01:07:39

1.5. Dynamic Programming

Bellman operator T: candidate continuation value $v \to$ updated value Tv.

$$(Tv)(k) = \max_{c,k'} [u(c) + \beta v(k')] \tag{1.12}$$

subject to

$$c + k' = f(k) + (1-\delta)k, c, k' \geq 0. \tag{1.13}$$

Bellman solution = **fixed point**:

$$Tv = v \Leftrightarrow (Tv)(k) = v(k) \text{ for all feasible } k. \tag{1.14}$$

Fixed-point theorems give existence/uniqueness conditions within chosen function space; assumptions matter.

1.6. Contraction mapping theorem

Theorem 1.6.1 (Contraction mapping theorem): Let (S, d) be a nonempty complete metric space and $T : S \to S$ a contraction: there exists $\beta \in (0, 1)$ such that

$$d(Tx, Ty) \leq \beta d(x, y) \text{ for all } x, y \in S. \tag{1.15}$$

Then T has a unique fixed point $v \in S$. For every $v_0 \in S$ and $n = 0, 1, 2, \ldots$,

$$d(T^n v_0, v) \leq \beta^n d(v_0, v). \tag{1.16}$$

Hence $T^n v_0 \to v$ as $n \to \infty$, with geometric convergence. Here T^n denotes n successive applications of T, with T^0 the identity.

Lecture addition: Intuition for Contraction Mapping

- Analogous to Brouwer fixed-point theorem in 1D: continuous function mapping interval to itself with slope $< 1 \to$ unique fixed point.
- In function spaces, "slope" = contraction factor β. Iterating operator moves points closer → convergence to unique fixed point.

Recording: 00:59:06, 01:01:46

1.6.1. Application to planner

- Assume feasible capital remains in nonempty compact interval K.
- $S = C(K)$: continuous real-valued functions on K; automatically bounded.
- **Sup-norm metric**:

$$d(v, w) = \|v - w\|_\infty = \sup_{k \in K} |v(k) - w(k)|. \tag{1.17}$$

  (S, d) complete.
- Must verify $T : S \to S$: updated values remain bounded and continuous. Then Bellman discounting gives

$$\|Tv - Tw\|_\infty \leq \beta \|v - w\|_\infty. \tag{1.18}$$

  Theorem 1.6.1 → unique fixed point in S; value iteration converges from any $v_0 \in S$.

Lecture addition: Economic Interpretation of VFI

- **Initial guess $v^0 = 0$**: One-period model (no future utility). Agent consumes all output today.
- **First iteration Tv^0**: Two-period model. Optimize today/tomorrow, world ends after tomorrow.
- **Convergence**: Each iteration extends horizon by 1 period. $n \to \infty$ → infinite-horizon value function.

Recording: 00:56:48, 00:58:31

1.7. Using the value function analytically

Assume u, f, and v differentiable; optimal choices interior. Substitute $c = f(k) + (1 - \delta)k - k'$ into Bellman:

$$v(k) = \max_{k'} [u(f(k) + (1 - \delta)k - k') + \beta v(k')], \tag{1.19}$$

over feasible k'.

- **FOC for k'**:

$$-u'(c) + \beta v'(k') = 0 \Rightarrow u'(c) = \beta v'(k'). \tag{1.20}$$

- **Envelope condition**: differentiate w.r.t. k, holding optimal k' fixed:

$$v'(k) = u'(c)[f'(k) + 1 - \delta]. \tag{1.21}$$

- **Combine**: apply envelope condition next period → usual Euler equation:

$$u'(c_t) = \beta v'(k_{t+1}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]. \tag{1.22}$$

  Same condition as sequence problem. At corners, use KKT conditions.

1.8. Using the value function numerically

Need numerical representation of $v(k)$.

- Simplest approach: restrict capital to finite grid

$$K_N = \{k^1, k^2, ..., k^N\} \subset K. \tag{1.23}$$

- Approximate value function by vector

$$\boldsymbol{v} = (v_1, v_2, ..., v_N)^\top \in \mathbb{R}^N, v_i \approx v(k^i). \tag{1.24}$$

One entry per capital grid point.

Algorithm 1.8.1 (Value function iteration (VFI)):

1. **Choose grid** $K_N = \{k^1, ..., k^N\}$. For each pair (i, j), compute

$$c_{ij} = f(k^i) + (1-\delta)k^i - k^j. \tag{1.25}$$

   Let J_i contain choices j with $c_{ij} \geq 0$ and finite $u(c_{ij})$. Each J_i must be nonempty.
2. **Initial guess**: any $\boldsymbol{v}^0 \in \mathbb{R}^N$, e.g. all zeros. Set iteration $m = 0$ and tolerance $\varepsilon > 0$.
3. **Bellman update**: for every $i = 1, ..., N$,

$$v_i^{m+1} = \max_{j \in J_i} \left[u(c_{ij}) + \beta v_j^m \right]. \tag{1.26}$$

   Use the old vector for all updates: $\boldsymbol{v}^{m+1} = T_N \boldsymbol{v}^m$.
4. **Convergence check**:

$$\Delta_m = \max_{1 \leq i \leq N} \left| v_i^{m+1} - v_i^m \right|. \tag{1.27}$$

   If $\Delta_m \leq \varepsilon$, stop and return $\hat{\boldsymbol{v}} = \boldsymbol{v}^{m+1}$. Otherwise set $m \leftarrow m + 1$ and repeat step 3.
5. **Recover approximate policy**: for each i, choose

$$j_i \in \arg\max_{j \in J_i} \left[u(c_{ij}) + \beta \hat{v}_j \right]. \tag{1.28}$$

   Set $k'(k^i) = k^{j_i}$ and $c(k^i) = c_{ij_i}$.

Why it converges: T_N is a β-contraction in the sup norm. Theorem 1.6.1 → unique grid fixed point $\boldsymbol{v}_N^*$; Algorithm 1.8.1 converges from any initial vector. At stopping, value-error bound:

$$\|\hat{\boldsymbol{v}} - \boldsymbol{v}_N^*\|_\infty \leq \frac{\beta}{1-\beta} \Delta_m \leq \frac{\beta}{1-\beta} \varepsilon. \tag{1.29}$$

This controls iteration error; grid approximation error remains.

Lecture addition: Convergence in General Equilibrium

- CMT guarantees convergence for individual decision problems (value functions).
- Heterogeneous GE: convergence of **distribution** of states (assets, income, human capital) required.
- Distribution = higher-dimensional object; may have multiple fixed points → more complex than single-agent VFI.

Recording: 01:14:16, 01:15:18

Lecture addition: Curse of Dimensionality

- Grid-based VFI suffers from **curse of dimensionality**: n state variables → M^n grid points (exponential growth).
- Large n (e.g., 50) → computationally intractable.
- **Frontier approaches**: Sparse grids; mean field games [ASR uncertain: Ben Moll]; heuristic trust in convergence when rigorous proofs unavailable.

Recording: 01:15:51, 01:16:53

Lecture addition: Complexity vs. Reality

- Model complexity driven by **expectation side**: agents anticipate others' actions → complex fixed-point problems.
- Evidence: real people not "fully forward-looking" as standard models assume.
- **Responses**: (1) Develop computational tools for complex models; (2) Simplify models to match cognitive complexity (behavioral/limited rationality).

Recording: 01:17:25, 01:19:00

2. Lecture 2 Oct 5

2.1. Recall

2.1.1. Bellman equation for the standard growth model

$$v(k) = \max_{c,k'}[u(c) + \beta v(k')], u(c) = \frac{c^{1-\sigma}}{1-\sigma}, \tag{2.1}$$

subject to

$$c + k' = (1-\delta)k + k^{\alpha}, c > 0, k' \geq 0. \tag{2.2}$$

For $\sigma = 1$, use $u(c) = \log c$.

2.1.2. Contraction mapping theorem

On a nonempty complete metric space (S, d), a contraction $T : S \to S$ with modulus $\beta \in (0, 1)$ has a unique fixed point v (Theorem 1.6.1):

$$\begin{aligned} d(Tx, Ty) &\leq \beta d(x, y), \\ d(T^n v_0, v) &\leq \beta^n d(v_0, v). \end{aligned} \tag{2.3}$$

Iteration from any $v_0 \in S$ converges geometrically to v.

2.2. Blackwell's sufficient conditions

Theorem 2.2.1 (Blackwell's sufficient conditions): Let $B(X)$ be the bounded real-valued functions on X, equipped with the sup norm, and let $T : B(X) \to B(X)$. Suppose:

1. **Monotonicity**: For all $f, g \in B(X)$,

$$f(x) \geq g(x) \text{ for all } x \in X \implies (Tf)(x) \geq (Tg)(x) \text{ for all } x \in X. \tag{2.4}$$

2. **Discounting**: There exists $\beta \in (0, 1)$ such that, for every $f \in B(X)$ and $a \geq 0$,

$$(T(f + a))(x) \leq (Tf)(x) + \beta a \text{ for all } x \in X. \tag{2.5}$$

   Here $(f + a)(x) = f(x) + a$.

Then T is a contraction with modulus β:

$$\|Tf - Tg\|_{\infty} \leq \beta \|f - g\|_{\infty}. \tag{2.6}$$

2.2.1. Verification for the Bellman operator

Maximization preserves pointwise inequalities, so $f \geq g \implies Tf \geq Tg$. Adding a constant to continuation utility gives

$$(T(f + a))(k) = \max_{c,k'}[u(c) + \beta f(k')] + \beta a = (Tf)(k) + \beta a. \tag{2.7}$$

Both conditions hold; the self-map requirement $T : B(X) \to B(X)$ must also be verified.

2.3. Blackwell's conditions for the standard model

Let $\alpha, \delta, \beta \in (0,1)$. Resources and consumption:

$$\begin{aligned} y(k) &= \underbrace{(1-\delta)k}_{\text{remaining capital}} + \underbrace{k^{\alpha}}_{\text{output}}, \\ c(k,k') &= y(k) - k'. \end{aligned} \tag{2.8}$$

The invariant state space satisfies

$$\begin{aligned} \bar{k} &= \delta^{-\frac{1}{1-\alpha}}, \\ X &= \left[0, \bar{k}\right], \\ 0 \le y(k) &\le y\left(\bar{k}\right) = \bar{k} \text{ for all } \; k \in X. \end{aligned} \tag{2.9}$$

> **Lecture addition: The capital ceiling**
>
> $\bar{k}$ is the steady state with zero consumption: all output goes into investment, and $\bar{k}^{\alpha} = \delta\bar{k}$. The bound comes from feasibility alone. Starting in X, feasible savings remain in X.
>
> Recording: 00:07:02 - 00:07:39

Feasible savings and the Bellman operator:

$$\begin{aligned} \Gamma(k) &= \{k' \ge 0 : c(k,k') \ge 0\} \subseteq X, \\ (Tf)(k) &= \sup_{k' \in \Gamma(k)} \left[\underbrace{u(c(k,k'))}_{\text{current utility}} + \beta \underbrace{f(k')}_{\text{continuation value}} \right]. \end{aligned} \tag{2.10}$$

Extend u to $c = 0$ by its limit, possibly $-\infty$. Replace sup with max when the optimum is attained.

2.3.1. Monotonicity

For $f, g \in B(X)$ with $f \ge g$ pointwise,

$$\begin{aligned} (Tf)(k) &= \sup_{k' \in \Gamma(k)} \left[u(c(k,k')) + \beta f(k')\right] \\ &\ge \sup_{k' \in \Gamma(k)} \left[u(c(k,k')) + \beta g(k')\right] \\ &= (Tg)(k). \end{aligned} \tag{2.11}$$

Lecture addition: Why the first inequality can be strict

If the maximum for g is attained, let $k^\star(k)$ be its optimal savings:

$$\begin{aligned}(Tf)(k) &\geq u(c(k,k^\star(k))) + \beta f(k^\star(k)),\\ &\geq u(c(k,k^\star(k))) + \beta g(k^\star(k)),\\ &= (Tg)(k).\end{aligned}$$

The first step evaluates one feasible choice. It can be strict because $k^\star(k)$ need not maximize the objective with continuation value f.

Recording: 00:08:00 - 00:10:59

2.3.2. Discounting

For $f \in B(X)$ and a constant $a \geq 0$,

$$\begin{aligned}(T(f+a))(k) &= \sup_{k' \in \Gamma(k)} \left[u(c(k,k')) + \beta f(k') + \underbrace{\beta a}_{\text{constant in } k'} \right] \\ &= (Tf)(k) + \beta a.\end{aligned} \tag{2.12}$$

2.3.3. Boundedness and conclusion

For $0 < \sigma < 1$, $u(0) = 0$ and

$$\begin{aligned}0 \leq u(c(k,k')) &\leq u\left(\overline{k}\right) < \infty,\\ \|Tf\|_\infty &\leq u\left(\overline{k}\right) + \beta \|f\|_\infty < \infty.\end{aligned} \tag{2.13}$$

Thus $T : B(X) \to B(X)$ on the complete sup-norm space $B(X)$. By Theorem 2.2.1 and Theorem 1.6.1, there is a unique $v \in B(X)$ with $Tv = v$, and for every $v_0 \in B(X)$,

$$\|T^n v_0 - v\|_\infty \leq \beta^n \|v_0 - v\|_\infty \to 0. \tag{2.14}$$

For $\sigma \geq 1$ (including log utility),

$$k = 0 \implies \Gamma(k) = \{0\} \implies (Tf)(0) = -\infty. \tag{2.15}$$

The bounded-function theorem then requires a different state or function space.

2.4. Formulating recursive models

Identify the state variables before writing the Bellman equation.

A **state variable**:
- Matters for the decision problem.
- Is taken as given by the decision maker at the time of choice.
- Varies over time or across states.

Other variables:
- **Choice variables**: chosen by the decision maker.

- **Parameters**: affect the problem but are fixed over time and across states.

The value function takes the state variables as its arguments.

> **Lecture addition: A past choice can be a current state**
>
> The household chose today's capital in the past; it is fixed when today's consumption-saving decision is made. Models with habits or varying patience may require preference states. Here β and σ remain parameters because they are fixed.
>
> Recording: 00:16:22 - 00:18:54

2.4.1. How to determine the state

1. **Fix timing**: Specify what is observed before choices are made. A lagged choice may be today's state; a newly chosen quantity is a control.
2. **List candidates**: Retain inherited stocks and observed shocks that affect current constraints, utility, or forecasts. Separate fixed parameters.
3. **Check sufficiency**: Given state s and choices q, specify

$$\begin{aligned} q &\in \Gamma(s), \\ \text{current utility} &= U(s,q), \\ s' &= G(s,q,\varepsilon'). \end{aligned} \tag{2.16}$$

   Here ε' is future uncertainty. Two histories with the same s must give the same feasible choices, current utility, and conditional distribution of s' for every feasible q.
4. **Remove redundancy**: Drop a coordinate only if this sufficiency test still holds. A one-to-one change of coordinates gives an equivalent state.
5. **Write the Bellman equation**:

$$v(s) = \sup_{q \in \Gamma(s)} \left[\underbrace{U(s,q)}_{\text{current utility}} + \beta \underbrace{\mathbb{E}[v(s') \mid s,q]}_{\text{expected continuation}} \right]. \tag{2.17}$$

   Use max when the optimum is attained; omit the expectation in a deterministic model.

Reduction test: If two candidate states have the same proposed summary but different feasible choices, utility, or transition distributions, the summary is insufficient.

2.4.2. Example: standard model with two sectors

Each period, allocate inherited capital k and one unit of labor freely between consumption (c) and investment (i) sectors. Let $\alpha, \gamma \in (0,1)$.

Feasibility:

$$
\begin{aligned}
c &= \underbrace{k_c^{\alpha} n_c^{1-\alpha}}_{\text{consumption output}}, \\
k' &= \underbrace{(1-\delta)k}_{\text{remaining capital}} + \underbrace{k_i^{\gamma} n_i^{1-\gamma}}_{\text{investment output}}, \\
k_c + k_i &= k, \\
n_c + n_i &= 1.
\end{aligned}
\tag{2.18}
$$

All quantities are nonnegative.

State identification:

- k is inherited; (k_c, k_i, n_c, n_i) are chosen before production.
- Free reallocation removes dependence on the previous sectoral split.
- Given k and current choices, utility and next-period k' are determined.

Hence $s = k$ is sufficient.

The Bellman equation is

$$
v(k) = \max_{c,k',k_c,k_i,n_c,n_i} [u(c) + \beta v(k')], \tag{2.19}
$$

subject to the constraints above, with u as defined in Recall.

> **Lecture addition: Timing and the sectoral split**
>
> Let $\tilde{v}(k_c, k_i)$ be the value after capital has been allocated across sectors. Before allocation,
>
> $$v(k) = \max_{k_c + k_i = k} \tilde{v}(k_c, k_i).$$
>
> In this deterministic model, the future split can be planned a period earlier. Treating the initial allocation consistently gives the same choices. With M grid points per state, retaining both stocks requires M^2 combinations; retaining total capital requires M.
>
> Recording: 00:24:30 - 00:30:44

2.4.3. Example: two sectors with irreversible investment

Capital is installed before the period and cannot move between sectors. Choose labor allocations and new investment x_c, x_i today; investment becomes productive next period.

Feasibility:

$$\begin{aligned} c &= k_c^{\alpha} n_c^{1-\alpha}, \\ x_c + x_i &= \underbrace{k_i^{\gamma} n_i^{1-\gamma}}_{\text{investment output}}, \\ k_c' &= (1-\delta)k_c + x_c, \\ k_i' &= (1-\delta)k_i + x_i, \\ n_c + n_i &= 1, \\ x_c, x_i &\geq 0. \end{aligned} \tag{2.20}$$

All other quantities are nonnegative. Irreversibility implies

$$k_j' \geq (1-\delta)k_j \text{ for } \; j \in \{c, i\}. \tag{2.21}$$

Sectoral capital can decline only through depreciation.

Lecture addition: Why the aggregate equation is insufficient

The state is (k_c, k_i). Adding the accumulation equations leaves the separate restrictions $x_c, x_i \geq 0$ in place. Dropping them would permit conversion of existing capital between sectors. They can bind when initial stocks are imbalanced or shocks change the desired sectoral mix.

Recording: 00:32:20 - 00:34:45

Lecture addition: Applications of multisector models

Separate consumption and investment sectors allow general and investment-specific technological change to have different effects on business cycles. Larger models use input-output networks, with sectors buying CES aggregates of intermediate inputs, to study shock propagation.

Recording: 00:20:42 - 00:20:53; 00:35:00 - 00:36:15

2.4.4. Standard model with shocks

Productivity z is i.i.d. over time. The resource constraint is

$$c + k' = (1-\delta)k + zk^{\alpha}. \tag{2.22}$$

In all formulations, $c, k' \geq 0$.

2.4.4.1. States (k, z): after the shock

Observe z, then choose c and k':

$$v(k, z) = \max_{c, k'} [u(c) + \beta \mathbb{E}_{z'}[v(k', z')]], \tag{2.23}$$

subject to Equation (2.22). The expectation is over next period's shock z'.

2.4.4.2. State x: total resources

After observing z, define

$$x = (1-\delta)k + zk^{\alpha}. \tag{2.24}$$

Then

$$w(x) = \max_{c,k'}[u(c) + \beta \mathbb{E}_{z'}[w(x')]], \tag{2.25}$$

subject to

$$\begin{aligned} c + k' &= x, \\ x' &= (1-\delta)k' + z'(k')^{\alpha}. \end{aligned} \tag{2.26}$$

z' is independent of z, so the transition of x' requires only k'.

Note: This reduction to x relies on z being i.i.d. over time.

> **Lecture addition: The forecast matters with persistent shocks**
>
> Two states can have the same current resources x but different productivity z. With persistence, their distributions of z' differ, so their optimal savings can differ. Current z carries information about tomorrow.
>
> Recording: 00:45:42 - 00:46:22

2.4.4.3. State k: before the shock

$V(k)$ is the value *before z is observed.* Choices $c(z)$ and $k'(z)$ may *depend on its realization:*

Chocies. *For a fixed (k, z) or x, Options 1 and 2 maximize over scalar choices c, k'. Their policy functions are $c(k, z), k'(k, z)$ or $c(x), k'(x)$; state arguments are implicit. Option 3 maximizes over functions $c(z), k'(z)$ for every possible z.*

$$V(k) = \max_{c(z),k'(z)} \mathbb{E}_z[u(c(z)) + \beta V(k'(z))], \tag{2.27}$$

subject to

$$c(z) + k'(z) = (1-\delta)k + zk^{\alpha} \text{ for every } z. \tag{2.28}$$

$$\begin{aligned} v(k, z) &= w((1-\delta)k + zk^{\alpha}), \\ V(k) &= \mathbb{E}_z[v(k, z)]. \end{aligned} \tag{2.29}$$

Timing. Before observing z, $V(k)$ averages over its realization; i.i.d. shocks make past productivity uninformative about z. After observing z, the state records realized resources through (k, z) or x. Decisions can depend on observed z in all three formulations.

Lecture addition: Several values within one period

A model can use value functions at several stages of a period. A value before information arrives is linked to the later value by an expectation over the information revealed between those stages.

Recording: 00:51:25 - 00:52:16

2.5. Equilibrium

A social planner maximizes welfare subject to feasibility constraints. In a decentralized economy, agents make their own choices, linked through markets, institutions, and policies.

- How can decentralized outcomes be represented formally?
- How do equilibrium allocations relate to the planner's solution?
- Can decentralized equilibria be formulated recursively?

2.6. What is a competitive equilibrium?

A competitive equilibrium consists of:
- An *allocation*: consumption, production, and factor inputs.
- A *price system*: prices in every market.

Two conditions:
- **Optimization**: Households maximize utility and firms maximize profits, taking prices as given.
- **Feasibility**: The allocation is feasible: total use of each good does not exceed production plus endowments.

Lecture addition: Trading dates and events

With complete markets, goods are distinguished by delivery date and event. The lecture's example is consumption in 2027 conditional on no rain in Australia. Suitable sequential asset markets can span the same contingent consumption trades as markets opened once at date 0.

Recording: 01:02:26 - 01:03:56

2.7. Sequential equilibrium

In the deterministic standard model, markets open each period. The household owns capital k_t and a fixed labor endowment $n_t = 1$. It rents capital and supplies labor to the firm; both agents take prices as given. Let r_t be the net return on capital. The capital rental rate is $r_t + \delta$; the wage is w_t.

Given k_0, a sequential equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{r_t, w_t\}_{t=0}^{\infty}$ satisfying the following conditions.

2.7.1. Household

Given the price paths, the household solves

$$\max_{\{c_t,k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{2.30}$$

subject to

$$\begin{aligned} c_t + k_{t+1} &= \underbrace{(1+r_t)k_t}_{\text{capital payoff}} + \underbrace{w_t n_t}_{\text{labor income}}, \\ n_t &= 1, \\ c_t &\geq 0, \\ k_{t+1} &\geq 0. \end{aligned} \tag{2.31}$$

2.7.2. Firm

The firm chooses capital demand K_t and labor demand N_t each period:

$$\max_{K_t,N_t\geq 0} \left[K_t^{\alpha} N_t^{1-\alpha} - (r_t+\delta)K_t - w_t N_t\right]. \tag{2.32}$$

Lecture addition: Representative agents and price taking

The representative firm behaves as an atomistic price taker. Its labor demand N_t is a choice; $N_t = 1$ is imposed through market clearing. Representative notation can stand for a continuum of identical agents. Separate demand and supply symbols make this distinction explicit.

Recording: 01:06:36 - 01:08:06; 01:12:24 - 01:13:53

2.7.3. Market clearing

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1-\delta)k_t + k_t^{\alpha}. \end{aligned} \tag{2.33}$$

At positive inputs, the firm's first-order conditions imply

$$\begin{aligned} r_t &= \alpha k_t^{\alpha-1} - \delta, \\ w_t &= (1-\alpha)k_t^{\alpha}. \end{aligned} \tag{2.34}$$

Lecture addition: Depreciation and the quoted return

An alternative convention quotes the capital rental rate $\rho_t = r_t + \delta$ and lets the household bear depreciation. Then

$$\begin{aligned} c_t + k_{t+1} &= (1-\delta+\rho_t)k_t + w_t n_t, \\ \text{firm capital cost} &= \rho_t K_t. \end{aligned}$$

The net-return and rental-rate conventions describe the same payments.

Recording: 01:10:41 - 01:11:28; 01:14:52 - 01:16:21

2.7.4. No-Ponzi condition

If borrowing is allowed, replace $k_{t+1} \geq 0$ with

$$\liminf_{t\to\infty} \frac{k_{t+1}}{\prod_{s=1}^{t}(1+r_s)} \geq 0. \tag{2.35}$$

At an interior optimum with $u'(c_s) > 0$, the Euler equation gives

$$1 + r_s = \frac{u'(c_{s-1})}{\beta u'(c_s)}. \tag{2.36}$$

Multiplying from $s = 1$ to t, marginal utilities cancel:

$$\begin{aligned} \prod_{s=1}^{t}(1+r_s) &= \prod_{s=1}^{t} \frac{u'(c_{s-1})}{\beta u'(c_s)}, \\ &= \frac{u'(c_0)}{\beta^t u'(c_t)}. \end{aligned} \tag{2.37}$$

Hence

$$\frac{k_{t+1}}{\prod_{s=1}^{t}(1+r_s)} = \frac{\beta^t u'(c_t) k_{t+1}}{u'(c_0)}. \tag{2.38}$$

Since $u'(c_0) > 0$ is constant, dividing by it preserves the sign of the limit inferior. Thus the no-Ponzi condition is equivalent to

$$\liminf_{t\to\infty} \beta^t u'(c_t) k_{t+1} \geq 0. \tag{2.39}$$

The household cannot finance consumption by rolling over debt forever.

> **Lecture addition: Physical capital and financial assets**
>
> Physical capital is nonnegative. An individual's financial asset position can be negative when borrowing is allowed for consumption smoothing. Period budgets alone permit arbitrarily large debt followed by refinancing of principal and interest. A borrowing limit or no-Ponzi condition restricts those plans.
>
> Recording: 01:16:28 - 01:17:20; 01:18:29 - 01:19:47

2.8. Arrow-Debreu equilibrium

Sequential markets open each period. In an Arrow-Debreu equilibrium, markets open once at date 0. Agents contract for goods delivered at each future date (and state, under uncertainty). Delivery and production still occur each period.

Given k_0, an Arrow-Debreu equilibrium consists of an allocation $\{c_t, n_t, k_{t+1}\}_{t=0}^{\infty}$ and prices $\{p_t, r_t, w_t\}_{t=0}^{\infty}$ satisfying the conditions below.

The additional price $p_t > 0$ is the date-0 price of one unit of the good delivered at date t, with $p_0 = 1$. It converts date-t quantities into date-0 goods.

The wage w_t and capital rental rate $r_t + \delta$ remain measured in date-t goods. Their date-0 prices are $p_t w_t$ and $p_t(r_t + \delta)$.

2.8.1. Household

The household solves

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{2.40}$$

subject to $n_t = 1$, $c_t \geq 0$, given k_0, and a single present-value budget:

$$\sum_{t=0}^{\infty} p_t(c_t + k_{t+1}) = \sum_{t=0}^{\infty} p_t[(1 + r_t)k_t + w_t n_t]. \tag{2.41}$$

The present-value sums are assumed finite.

2.8.2. Firm and market clearing

For each delivery date t, the firm solves

$$\max_{K_t, N_t \geq 0} \left[K_t^{\alpha} N_t^{1-\alpha} - (r_t + \delta)K_t - w_t N_t\right]. \tag{2.42}$$

Multiplying profits by $p_t > 0$ leaves the input choices unchanged. Markets clear at every date:

$$\begin{aligned} K_t &= k_t, \\ N_t &= n_t = 1, \\ c_t + k_{t+1} &= (1 - \delta)k_t + k_t^{\alpha}. \end{aligned} \tag{2.43}$$

2.8.3. No arbitrage and the present-value budget

Definition 2.8.1 (No arbitrage in capital): In this deterministic Arrow-Debreu economy, one unit of k_{t+1} costs p_t at date 0 and pays $p_{t+1}(1 + r_{t+1})$ in date-0 goods. No arbitrage in capital requires

$$\begin{aligned} p_t - p_{t+1}(1 + r_{t+1}) &= 0, \\ \frac{p_t}{p_{t+1}} &= 1 + r_{t+1}. \end{aligned} \tag{2.44}$$

Thus $\frac{p_t}{p_{t+1}}$ is the gross real return from t to $t + 1$.

Lecture addition: Why positive capital requires no arbitrage

One extra unit of k_{t+1} changes lifetime wealth by

$$p_{t+1}(1 + r_{t+1}) - p_t.$$

- If negative, the household chooses $k_{t+1} = 0$.
- If positive, capital purchases generate unbounded wealth; there is no finite optimum.
- If zero, the household is indifferent over capital holdings at these prices.

An equilibrium with positive capital therefore requires equality. Future capital purchases and payoffs cancel; **initial capital** remains part of wealth because its acquisition cost is outside the lifetime budget.

Recording: 2026-10-06, 00:00:46 - 00:03:31

With $p_0 = 1$,

$$p_t = \frac{1}{\prod_{s=1}^{t}(1 + r_s)}. \tag{2.45}$$

Capital purchases and their future payoffs cancel from the lifetime budget:

$$\sum_{t=0}^{\infty} p_t c_t = \underbrace{(1 + r_0)k_0}_{\text{initial capital wealth}} + \underbrace{\sum_{t=0}^{\infty} p_t w_t}_{\text{lifetime labor income}}. \tag{2.46}$$

The lifetime budget makes a separate no-Ponzi condition unnecessary.

Lecture addition: Valuing other assets

Date prices also value a claim to a dividend stream d_t:

$$q_0 = \sum_{t=1}^{\infty} p_t d_t.$$

The same method prices a bond by multiplying each payoff by its delivery-date price. With uncertainty, use **state-contingent prices**. This is why Arrow-Debreu notation is convenient for asset pricing.

Recording: 2026-10-06, 00:04:08 - 00:05:13

2.9. Recursive equilibrium

Recursive methods apply to **decentralized economies** as well as planning problems. Build on **sequential equilibrium**: agents choose each period, and state variables evolve over time.

In the standard model, distinguish two states:
- k: **individual capital**, which determines household wealth.
- K: **aggregate capital**, which determines prices and their future path.

The household chooses k' taking $K' = g(K)$ and the associated prices *as given*. In equilibrium, $k = K$, but keeping both states in the household problem preserves **price-taking**. With heterogeneity, individual and aggregate states generally differ.

Lecture addition: Whose saving moves tomorrow's prices?

Think of the representative household as one of many identical, tiny households. More **aggregate** saving lowers tomorrow's return to capital and raises wages. One household's saving has no effect on either price.

Thus $K' = g(K)$ is taken as given inside the household problem. Impose $g(K) = k'(K, K)$ when checking equilibrium consistency.

Recording: 2026-10-06, 00:07:30 - 00:08:48; 00:11:18 - 00:11:46

2.9.1. Standard model

A recursive competitive equilibrium consists of:

- **Value function:** $v(k, K)$.
- **Policy functions:** $c(k, K)$ and $k'(k, K)$.
- **Price functions:** $r(K)$ and $w(K)$.
- **Aggregate law of motion:** $g(K)$.

These functions satisfy four conditions:

1. **Household optimality.** Given r, w and g, the value function solves

$$v(k, K) = \max_{c, k' \geq 0} \{u(c) + \beta v(k', K')\}, \tag{2.47}$$

   subject to

$$\begin{aligned} c + k' &= (1 + r(K))k + w(K), \\ K' &= g(K). \end{aligned} \tag{2.48}$$

   The policies $c(k, K)$ and $k'(k, K)$ attain this maximum.

2. **Firm optimality.** With aggregate labor equal to one and $K > 0$,

$$\begin{aligned} r(K) &= \alpha K^{\alpha - 1} - \delta, \\ w(K) &= (1 - \alpha) K^{\alpha}. \end{aligned} \tag{2.49}$$

   Here $r(K)$ is the *net return*; the capital rental rate is $r(K) + \delta$.

3. **Market clearing.** For every admissible K,

$$c(K, K) + k'(K, K) = (1 - \delta)K + K^{\alpha}. \tag{2.50}$$

4. **Consistency.** The *perceived* aggregate law must equal the law *generated* by optimal household saving:

$$g(K) = k'(K, K). \tag{2.51}$$

2.10. The First Welfare Theorem

Theorem 2.10.1 (First Welfare Theorem): With **locally nonsatiated preferences**, **complete competitive markets** and **no externalities**, every competitive equilibrium allocation is **Pareto efficient**. This formulation assumes finitely many consumers and well-defined lifetime budgets.

Pareto efficiency means that no feasible allocation makes everyone weakly better off and at least one person strictly better off.

2.10.1. Key assumptions

- **Local nonsatiation:** every neighborhood of a consumption bundle contains a strictly preferred bundle.
- **Complete markets:** all relevant goods, delivery dates and states can be traded.
- **No externalities:** effects on others operate through market prices and trades.

Convexity is not required. Economies with *infinitely many overlapping generations* lie outside this finite-consumer result; competitive equilibria can be inefficient.

Lecture addition: Why overlapping generations are an exception

In a two-period OLG economy, infinitely many generations can trade an intrinsically worthless asset. The young give goods to the old for money, then sell that money to the next young generation when they retire. Under suitable conditions, this transfer scheme can improve on a nonmonetary equilibrium.

A last generation breaks the argument: it cannot sell the money onward and loses from accepting it. The infinite sequence of consumers matters for this exception to the First Welfare Theorem.

Recording: 2026-10-06, 00:18:10 - 00:21:32

2.10.2. Proof by contradiction of Theorem 2.10.1

Proof: Let $p \geq 0$ denote equilibrium prices and ω aggregate endowments. At equilibrium, consumer i consumes $x_i^\star$ and has wealth m_i from endowments and distributed profits; firm j produces $y_j^\star$.

Suppose $\left(\tilde{x}_i, \tilde{y}_j\right)$ is a feasible Pareto improvement: every consumer weakly prefers $\tilde{x}_i$ to $x_i^\star$, and consumer h strictly prefers $\widetilde{x_h}$ to $x_h^\star$.

1. **Consumer optimality.**
   - **Strict improvement:** the bundle must cost more than m_i; otherwise it was affordable.
   - **Weak improvement:** the bundle cannot cost less than m_i. With *budget slack*, local nonsatiation gives a nearby, strictly better bundle that remains affordable.

Either violation contradicts optimality. Hence

$$\begin{aligned} &p \cdot \tilde{x}_i \geq m_i \text{ for every } i, \\ &p \cdot \widetilde{x_h} > m_h \text{ for the strictly improved consumer } h. \end{aligned} \tag{2.52}$$

Summing gives

$$\sum_i p \cdot \tilde{x}_i > \sum_i m_i. \tag{2.53}$$

2. **Feasibility and firm optimality.** Consumption is bounded by endowments plus production.

   Profit maximization implies $p \cdot \tilde{y}_j \leq p \cdot y_j^\star$ for each firm. Therefore

$$\begin{aligned} \sum_i p \cdot \tilde{x}_i &\leq p \cdot \omega + \sum_j p \cdot \tilde{y}_j \text{ (feasibility)}, \\ &\leq p \cdot \omega + \sum_j p \cdot y_j^\star \text{ (profit maximization)}, \\ &= \sum_i m_i \text{ (endowments + distributed profits)}. \end{aligned} \tag{2.54}$$

3. **Contradiction.** Total cost cannot be both *greater than* and *at most* aggregate wealth. No feasible Pareto improvement exists.

□

2.10.3. Implication for the standard model

In the **representative-agent standard model**, a competitive equilibrium allocation solves the planner's problem. *If the planner's solution is unique and equilibrium exists*, the allocations coincide. Recover prices from firms' marginal products.

2.11. The Second Welfare Theorem

Theorem 2.11.1 (Second Welfare Theorem): With complete competitive markets, no externalities and the conditions below, any **Pareto-efficient allocation** can be supported as a **competitive equilibrium** after **lump-sum wealth transfers**.

2.11.1. Sufficient conditions

- **Consumers:** a finite set.
- **Consumption:** each consumption set X_i is convex.
- **Preferences:** u_i is continuous and locally nonsatiated. For $x, x' \in X_i$ and $0 < \lambda < 1$, assume

$$u_i(x) > u_i(x') \Longrightarrow u_i(\lambda x + (1-\lambda)x') > u_i(x'). \tag{2.55}$$

- **Production:** the aggregate production set $Y = \sum_j Y_j$ is convex.
- **Separation:** the commodity space S is finite-dimensional, or Y has a nonempty interior.

- **Budget regularity:** at supporting prices p, each consumer has a feasible bundle $x_i^0 \in X_i$ with

$$p \cdot x_i^0 < p \cdot x_i^\star. \tag{2.56}$$

Convexity alone guarantees only a *quasi-equilibrium*: a strictly preferred bundle may cost exactly the consumer's wealth.[1]

2.11.2. Wealth transfers

At supporting prices, m_i includes original endowments and profits at the target production plan.

For target consumption $x_i^\star$, choose **transfers** T_i:

$$\begin{aligned} T_i &= \underbrace{p \cdot x_i^\star}_{\text{target wealth}} - \underbrace{m_i}_{\text{original wealth}}, \\ p \cdot x_i &\le m_i + T_i = p \cdot x_i^\star, \\ \sum_i T_i &= 0. \end{aligned} \tag{2.57}$$

Feasibility and profit distribution imply $\sum_i p \cdot x_i^\star = \sum_i m_i$, so transfers sum to zero. **Lump-sum** transfers are *independent of choices*, leaving marginal incentives unchanged.

2.11.3. Supporting prices

1. **Separation:** convexity permits a separating hyperplane between feasible aggregate resources and aggregate bundles that improve welfare.
2. **Prices:** the separating functional defines the price system p.
3. **Optimization:** firms maximize profits at these prices; budget regularity ensures consumers optimize their transfer-adjusted budgets.

2.12. How do the welfare theorems matter?

- **Markets and information:** under the welfare-theorem assumptions, equilibrium prices coordinate decentralized choices into an efficient allocation. *The Use of Knowledge in Society* (Hayek, 1945) explains how prices communicate *dispersed information*.
- **Policy:** diagnose failed assumptions, such as externalities, market power or missing markets. Under the second theorem's conditions, *lump-sum redistribution* can implement different efficient distributions without changing marginal incentives.
- **Computing equilibrium:** in the standard model, solve the planner's Bellman equation, then recover prices from marginal products. With heterogeneity, a planner allocation may require *wealth transfers* to be decentralized.

[1] See Condie's welfare-theorem notes, Proposition 6.1 ° for the budget condition. In an infinite-dimensional commodity space, $p \cdot x$ denotes the continuous price functional applied to x.

Lecture addition: Diagnosing a policy problem

Ask which welfare-theorem assumption fails in the proposed application.

- **Pollution:** an upstream producer harms downstream users without paying for that harm. A market for the relevant rights can price the externality.
- **Health insurance:** adverse selection can prevent useful insurance markets from forming. A missing market may justify intervention, but the underlying information problem still needs to be addressed.

Recording: 2026-10-06, 00:27:40 - 00:29:30

Lecture addition: What should an economist remember?

The class proposed equilibrium, Nash equilibrium, supply and demand, and economy-wide resource constraints. **Nash equilibrium** concerns individual best responses; maximizing total welfare is a different condition.

Economics often starts from **purposeful choices**: agents have objectives and face constraints. Macroeconomics asks how these choices interact and add up. Giving one group more resources requires accounting for where those resources come from.

Recording: 2026-10-06, 00:32:07 - 00:39:19

3. Lecture 4 Oct 6

3.1. Fundamental models of economic growth

The **standard model** is a Solow growth model with *optimizing consumers*. It matches long-run growth facts for high-income countries, but growth models should also explain:

- Economic development beyond the period described by the Kaldor facts.
- Income and growth differences across *all countries*.
- The origin of productivity growth.

Definition 3.1.1 (Kaldor's growth facts): Kaldor's six **stylized facts** (Kaldor, 1961) summarize long-run growth patterns, originally observed in industrialized economies:

1. **Output per worker** $\frac{Y_t}{N_t}$ grows at a roughly constant positive rate.
2. **Capital per worker** $\frac{K_t}{N_t}$ grows over time.
3. The **return to capital** is roughly constant.
4. The **capital-output ratio** $\frac{K_t}{Y_t}$ is roughly constant.
5. **Labor and capital income shares** are roughly constant.
6. **Output-per-worker growth rates** differ substantially across countries.

3.2. Solow

3.2.1. Setup

Output combines capital K_t and effective labor $A_t N_t$:

$$Y_t = K_t^{\alpha}(A_t N_t)^{1-\alpha}, 0 < \alpha < 1. \tag{3.1}$$

Here N_t is population (and labor supply); A_t is **labor-augmenting productivity**. Both grow *exogenously*:

$$\begin{aligned} A_{t+1} &= (1+g)A_t, \\ N_{t+1} &= (1+n)N_t. \end{aligned} \tag{3.2}$$

Assume $g, n \geq 0$. A **fixed saving rate** $0 < s < 1$ determines investment:

$$\begin{aligned} K_{t+1} &= (1-\delta)K_t + sY_t, \\ C_t &= (1-s)Y_t, \end{aligned} \tag{3.3}$$

where $0 \leq \delta \leq 1$ is depreciation. In the *standard model*, households instead choose saving optimally.

3.2.2. Capital per effective worker

Normalize by effective labor:

$$
\begin{aligned}
k_t &= \frac{K_t}{A_t N_t}, \\
y_t &= \frac{Y_t}{A_t N_t} = k_t^{\alpha}, \\
c_t &= \frac{C_t}{A_t N_t} = (1-s)k_t^{\alpha}.
\end{aligned}
\tag{3.4}
$$

Divide capital accumulation by $A_t N_t$ and use $A_{t+1}N_{t+1} = (1+g)(1+n)A_t N_t$:

$$
\begin{aligned}
k_{t+1}(1+g)(1+n) &= (1-\delta)k_t + sk_t^{\alpha}, \\
k_{t+1} - k_t &= \frac{sk_t^{\alpha} - (\delta + g + n + gn)k_t}{(1+g)(1+n)}.
\end{aligned}
\tag{3.5}
$$

Effective capital rises when investment exceeds depreciation and the capital needed to keep up with population and productivity growth. The term gn is retained because time is *discrete*.

3.2.3. Balanced growth path

On a **balanced growth path (BGP)**, capital per effective worker is constant:

$$
\begin{aligned}
s\bar{k}^{\alpha} &= (\delta + g + n + gn)\bar{k}, \\
\bar{k} &= \left(\frac{s}{\delta + g + n + gn}\right)^{\frac{1}{1-\alpha}}.
\end{aligned}
\tag{3.6}
$$

This positive steady state requires $\delta + g + n + gn > 0$. For $k_0 > 0$, diminishing returns imply convergence: k_t rises below $\bar{k}$ and falls above it.

- **Aggregate quantities:** K_t, Y_t and C_t grow at rate $(1+g)(1+n) - 1$.
- **Per capita quantities:** $\frac{K_t}{N_t}$, $\frac{Y_t}{N_t}$ and $\frac{C_t}{N_t}$ grow at rate g.
- **Growth facts:** the capital-output ratio, factor income shares and the marginal product of capital are constant; real wages grow at rate g.

A higher saving rate raises the BGP **level** of income per capita, not its **growth rate** g. With $g > 0$, the BGP matches Kaldor's first five facts. Cross-country differences in long-run growth require different *exogenous* productivity growth rates g.

Lecture addition: Growth during the transition

Constant $k_t = \frac{K_t}{A_t N_t}$ does not mean constant K_t: on the BGP, aggregate capital grows with effective labor.

Cobb-Douglas **factor income shares** are constant even outside the BGP. Starting below $\overline{k}$, k_t rises, so

$$\frac{K_t}{Y_t} = k_t^{1-\alpha} \text{ rises,}$$

$$\text{MPK}_t = \alpha k_t^{\alpha-1} \text{ falls.}$$

Constant capital-output ratios and returns are properties of the BGP, rather than every point along the transition.

Recording: 2026-10-06, 00:46:30 - 00:46:49; 00:48:04 - 00:48:30

Lecture addition: What the fixed saving rate assumes

The Solow model imposes the consumption-saving choice through s. It is a **reduced-form behavioral assumption**. The standard model derives this choice from household optimization.

A steady ratio in effective units still permits growth in levels and transitional dynamics. The version here is deterministic because it contains no shocks.

Recording: 2026-10-06, 00:50:44 - 00:52:16

3.2.4. Limitations

- Persistent cross-country income gaps require *exogenous differences* in productivity, saving or population growth.
- Baseline calibrations can imply faster convergence than observed, even among high-income countries.
- Productivity growth is unexplained; the model also lacks a mechanism for the historical transition from long-run stagnation to sustained growth.

3.3. Malthus

3.3.1. Setup

The model describes a preindustrial economy with population N_t and **fixed land** $X > 0$:

$$Y_t = N_t^{\alpha}(A_t X)^{1-\alpha}, 0 < \alpha < 1. \tag{3.7}$$

Productivity grows exogenously at rate $g \geq 0$:

$$A_{t+1} = (1+g)A_t. \tag{3.8}$$

Population growth responds to income per capita:

$$\begin{aligned} \frac{N_{t+1}}{N_t} &= s\frac{Y_t}{N_t}, \\ N_{t+1} &= sY_t. \end{aligned} \tag{3.9}$$

Here $s > 0$ is a demographic parameter. Higher income raises fertility or reduces mortality. Population increases when $s\frac{Y_t}{N_t} > 1$.

> **Lecture addition: Two routes from income to population**
>
> Higher income can raise demand for children or improve survival through better nutrition, especially in infancy. Either route gives a positive income-population feedback.
>
> Unlike Solow's saving rate, this s is **not a fraction of output saved**; the model does not require $s < 1$. Its demographic interpretation differs even though it occupies the same place in the accumulation equation.
>
> Recording: 2026-10-06, 00:57:22 - 00:59:04; 01:00:08 - 01:00:14

3.3.2. Population per effective land unit

Normalize by effective land $A_t X$:

$$\begin{aligned} n_t &= \frac{N_t}{A_t X}, \\ \frac{Y_t}{A_t X} &= n_t^{\alpha}, \\ \frac{Y_t}{N_t} &= n_t^{\alpha-1}. \end{aligned} \tag{3.10}$$

Using $N_{t+1} = sY_t$ gives

$$\begin{aligned} n_{t+1}(1+g) &= sn_t^{\alpha}, \\ n_{t+1} &= \frac{s}{1+g} n_t^{\alpha}. \end{aligned} \tag{3.11}$$

Since $\alpha < 1$, income per person falls as population grows relative to effective land.

3.3.3. Balanced growth path

On the BGP, $n_t = \bar{n}$ is constant:

$$\begin{aligned} \bar{n}(1+g) &= s\bar{n}^{\alpha}, \\ \bar{n} &= \left(\frac{s}{1+g}\right)^{\frac{1}{1-\alpha}}. \end{aligned} \tag{3.12}$$

For $n_0 > 0$, n_t converges to $\bar{n}$: it rises below the steady state and falls above it.

BGP income per capita and population density are

$$\begin{aligned} \frac{Y_t}{N_t} &= \bar{n}^{\alpha-1} = \frac{1+g}{s}, \\ \frac{N_t}{X} &= A_t \bar{n}. \end{aligned} \tag{3.13}$$

- Population and aggregate output grow at rate g.
- **Income per capita is constant**, even with positive productivity growth.
- Holding s and g fixed, higher productivity levels imply greater population density and the same BGP income per capita.

A permanent increase in the *level* of productivity initially raises income per person. Higher income accelerates population growth. As population rises relative to land, diminishing returns bring income per person back toward $\frac{1+g}{s}$.

Lecture addition: Lower fertility raises levels, not long-run growth

Holding g fixed, a lower demographic parameter s raises BGP income per person, $\frac{1+g}{s}$, while per capita growth remains zero. The lecture used **later marriage** as an example: fewer childbearing years can lower fertility at a given income and support higher living standards within the Malthusian regime.

Recording: 2026-10-06, 01:07:11 - 01:09:19

Lecture addition: What if land expands?

Fixed land creates diminishing returns to population. If land instead grows exogenously, the same normalization uses $A_t X_t$. With $\frac{X_{t+1}}{X_t} = 1 + g_X$, effective land grows at

$$1 + g_{AX} = (1+g)(1+g_X).$$

Replacing g by g_{AX} gives BGP income per person $\frac{1+g_{AX}}{s}$. Frontier expansion can raise this **level** and sustain faster population growth; a constant expansion rate still leaves long-run income per person constant.

Recording: 2026-10-06, 01:09:24 - 01:11:28

- Explains the world until 1800s

3.3.4. Malthus versus Solow

Solow accumulates **capital**; Malthus accumulates **population**. On the BGP, the return to the accumulated factor is constant. With competitive factor pricing, Malthus implies

$$\begin{aligned} w_t &= \alpha \frac{Y_t}{N_t} = \alpha \frac{1+g}{s}, \\ r_t^X &= (1-\alpha)\frac{Y_t}{X}, \end{aligned} \tag{3.14}$$

where r_t^X is the rental rate of land. *Wages are constant and land rents grow at rate g.*

In Solow, the return to capital is constant and wages grow at rate g.

3.3.5. From Malthus to Solow

Malthus describes preindustrial stagnation; Solow describes sustained growth in modern economies. Neither model alone explains the transition.

Lecture addition: What must change to escape stagnation?

The lecture outlined three mechanisms for the next class:

- **Structural change:** move production from land-dependent agriculture toward industry and services, whose capital and labor inputs can expand.
- **Demographic change:** weaken the positive income-fertility relationship; richer modern economies generally have lower birth rates.
- **Productivity growth:** explain why it accelerates and how it interacts with the other mechanisms.

A student also raised the scale effect in endogenous growth: more people can produce more ideas. The lecturer noted that this is another mechanism studied in the literature.

Recording: 2026-10-06, 01:13:28 - 01:15:52

Malthus to Solow (Hansen & Prescott, 2002) combines both technologies in one economy. Firms choose which to operate as productivity grows.

3.3.5.1. Technologies and resources

Both technologies are available in every period and produce the **same good**:

$$\begin{aligned} Y_{M,t} &= A_{M,t} K_{M,t}^{\alpha_1} N_{M,t}^{\alpha_2} X^{1-\alpha_1-\alpha_2}, \\ Y_{S,t} &= A_{S,t} K_{S,t}^{\theta} N_{S,t}^{1-\theta}. \end{aligned} \tag{3.15}$$

The Malthus sector M uses land; the Solow sector S uses only capital and labor. Assume $\alpha_1, \alpha_2 > 0$, $\alpha_1 + \alpha_2 < 1$ and $0 < \theta < 1$. Sectoral **TFP** grows exogenously:

$$A_{j,t+1} = \gamma_j A_{j,t}, \quad j \in \{M, S\}. \tag{3.16}$$

Here $\gamma_M \geq 1$ and $\gamma_S > 1$ are gross growth factors. Competitive firms allocate capital and labor freely across sectors:

$$\begin{aligned} K_t &= K_{M,t} + K_{S,t}, \\ N_t &= N_{M,t} + N_{S,t}, \\ X &= 1. \end{aligned} \tag{3.17}$$

One period represents a generation, so capital depreciates fully ($\delta = 1$). Land does not depreciate. Aggregate feasibility is

$$C_t + K_{t+1} = Y_{M,t} + Y_{S,t}. \tag{3.18}$$

3.3.5.2. Households and population

Households live for two periods and work only when young. A household born at t chooses positive consumption and nonnegative capital k_{t+1} and land x_{t+1}. With $\beta > 0$, its objective is

$$\max \log c_{1,t} + \beta \log c_{2,t+1}, \tag{3.19}$$

subject to

$$\begin{aligned} c_{1,t} + k_{t+1} + q_t x_{t+1} &= w_t, \\ c_{2,t+1} &= r^K_{t+1} k_{t+1} + \left(r^X_{t+1} + q_{t+1}\right) x_{t+1}. \end{aligned} \tag{3.20}$$

Here q_t is the land price and r^K_t, r^X_t are capital and land rental rates. With full depreciation, r^K_{t+1} is the entire payoff from one unit of capital saved at t.

There are N_t young households at t. Population follows a **calibrated** function G:

$$N_{t+1} = G\left(c_{1,t}\right) N_t. \tag{3.21}$$

G rises with consumption at low living standards and falls at high living standards. Each young household supplies one unit of labor. Asset and goods markets clear when

$$\begin{aligned} K_{t+1} &= N_t k_{t+1}, \\ 1 &= N_t x_{t+1}, \\ C_t &= N_t c_{1,t} + N_{t-1} c_{2,t}. \end{aligned} \tag{3.22}$$

With interior asset holdings, household optimality gives

$$\begin{aligned} c_{1,t} &= \frac{w_t}{1+\beta}, \\ q_t r^K_{t+1} &= r^X_{t+1} + q_{t+1}, \\ K_{t+1} &= N_t \frac{\beta}{1+\beta} w_t - q_t. \end{aligned} \tag{3.23}$$

The second equation equates the returns on capital and land. The third subtracts land purchases from the young generation's total saving.

3.3.5.3. Solow sector entry

Normalize the output price to one. At given factor prices, the minimum cost of producing one unit with the Solow technology is

$$\frac{1}{A_{S,t}} \left(\frac{r^K_t}{\theta}\right)^{\theta} \left(\frac{w_t}{1-\theta}\right)^{1-\theta}. \tag{3.24}$$

To test entry, evaluate this cost at the **Malthus-only** factor prices $r^K_{M,t}$ and $w_{M,t}$. Entry is profitable when

$$A_{S,t} > \underbrace{\left(\frac{r^K_{M,t}}{\theta}\right)^{\theta}\left(\frac{w_{M,t}}{1-\theta}\right)^{1-\theta}}_{\text{entry threshold}}. \tag{3.25}$$

Once entry occurs, factor prices adjust. If both sectors operate, their marginal products of capital and labor must coincide:

$$\begin{aligned} r^K_t &= \alpha_1 \frac{Y_{M,t}}{K_{M,t}} = \theta \frac{Y_{S,t}}{K_{S,t}}, \\ w_t &= \alpha_2 \frac{Y_{M,t}}{N_{M,t}} = (1-\theta)\frac{Y_{S,t}}{N_{S,t}}. \end{aligned} \tag{3.26}$$

The Malthus sector remains active at every finite date. With fixed land, scaling its capital and labor down together makes their marginal products arbitrarily large, so allocating a small amount to this sector remains worthwhile.

3.3.5.4. Transition

1. Initially, only the Malthus technology operates. Population growth absorbs productivity gains, keeping living standards constant. Solow productivity continues to grow while that sector is unused.
2. On the Malthusian growth path, wages and capital rental rates are constant, so the entry threshold is fixed. Rising $A_{S,t}$ eventually crosses it, and firms allocate capital and labor to the Solow sector.
3. The Solow sector has constant returns in capital and labor, with no fixed land input. In the calibrated economy, its share expands and income per person grows. The Malthus sector's share tends to zero. Population growth eventually falls through the assumed function G.

3.3.5.5. Limitations

- Productivity growth remains exogenous.
- The demographic transition comes from the calibrated function G; fertility and mortality decisions are not derived from household choices.
- The two technologies produce the same good. Requiring a separate agricultural good for subsistence would change the mechanism.

4. Lecture 4 Oct 12

4.1. Consumption, heterogeneity and policy

Lecture addition: Consumption functions and consumption smoothing

In the simple Keynesian example, consumption responds mechanically to current income. An illustrative marginal propensity to consume of 0.8 gives

$$\text{multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{1 - 0.8} = 5.$$

An optimizing household instead considers lifetime resources:

- A permanent income gain supports a larger increase in consumption.
- A one-off windfall is spread across future periods through saving.
- An announced future income change can affect consumption today.

The saving-and-borrowing example assumes **incomplete insurance**. With complete markets, idiosyncratic windfalls would be shared through state-contingent claims.

Recording: 2026-10-12, 00:08:21 - 00:16:02

Lecture addition: Hall's consumption test

Consumption already incorporates information about expected future income. Hall's test asks whether other current information helps forecast its next change (Hall, 1978).

For the exact random-walk benchmark, assume an unconstrained interior choice, a constant gross return R, $\beta R = 1$ and quadratic utility $u'(c) = a - bc$, with $b > 0$. The Euler equation gives

$$\begin{aligned} u'(c_t) &= \beta R E_t[u'(c_{t+1})], \\ a - bc_t &= a - bE_t[c_{t+1}], \\ E_t[c_{t+1} - c_t] &= 0. \end{aligned}$$

Quadratic utility gives certainty equivalence. Precautionary saving can break the exact random-walk result.

Paper detail: Hall used aggregate US data. Current income did not help predict consumption once current consumption was controlled for, but stock prices did. The original empirical conclusion was therefore qualified.

Recording: 2026-10-12, 00:16:03 - 00:22:07

Lecture addition: From New Keynesian models to HANK

Woodford combines optimizing households with nominal price or wage rigidities (Woodford, 2003). Interest rates and expected future rates affect consumption through households' Euler equations.

Households differ in wealth, liquidity and debt exposure. A household facing mortgage repricing can react differently from a wealthy saver. **HANK** (Heterogeneous Agent New Keynesian) models incorporate these differences; the paper discussed was *Monetary Policy According to HANK* (Kaplan et al., 2018).

Recording: 2026-10-12, 00:24:00 - 00:31:09

Lecture addition: Debt deflation

With fixed nominal debt D, a fall in the price level raises its real value:

$$P' = \frac{P}{2} \quad \Rightarrow \quad \frac{D}{P'} = 2\frac{D}{P}.$$

Bankruptcies can disrupt production and access to finance. The lecturer suggested this could be an important channel behind severe downturns, possibly more important than a household demand shortfall. A related model of firm borrowing and collateral is *Credit Cycles* (Kiyotaki & Moore, 1997).

Recording: 2026-10-12, 00:32:24 - 00:34:35

Lecture addition: Ricardian equivalence as a benchmark

Hold government spending fixed. Debt finance postpones taxes. If households are unconstrained, forward looking and bear the future tax liabilities, changing the timing of taxes leaves lifetime wealth unchanged (Barro, 1974).

Borrowing constraints, finite horizons and limited foresight can break this neutrality. The benchmark helps identify which assumptions make the timing of taxes matter.

Recording: 2026-10-12, 00:34:45 - 00:37:03

4.2. Facts on growth and demographics

Figures from Lecture 5, slides 4–7.

- Income and population data: Maddison Project Database 2023 (Bolt & van Zanden, 2024).
- Fertility data: UN World Population Prospects (United Nations Population Division, 2024), via Our World in Data°.

4.2.1. Income per capita over the long run

Income remains low for centuries, then takes off. Takeoff dates differ across countries; modern income gaps are large.

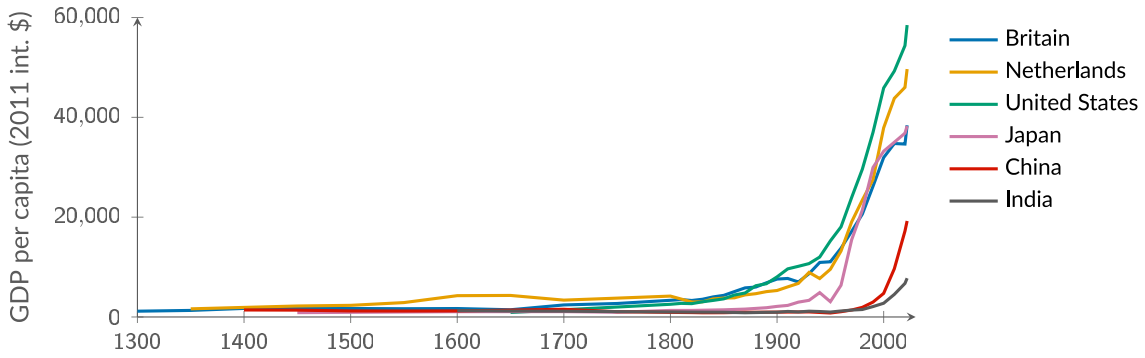


Figure 1: Income per capita over the long run.

Note: $y_t = \frac{Y_t}{N_t}$: income per capita in 2011 international dollars.

4.2.2. Population growth

Population growth has a hump shape in many regions, with peaks at different dates. Births, deaths and migration all matter.

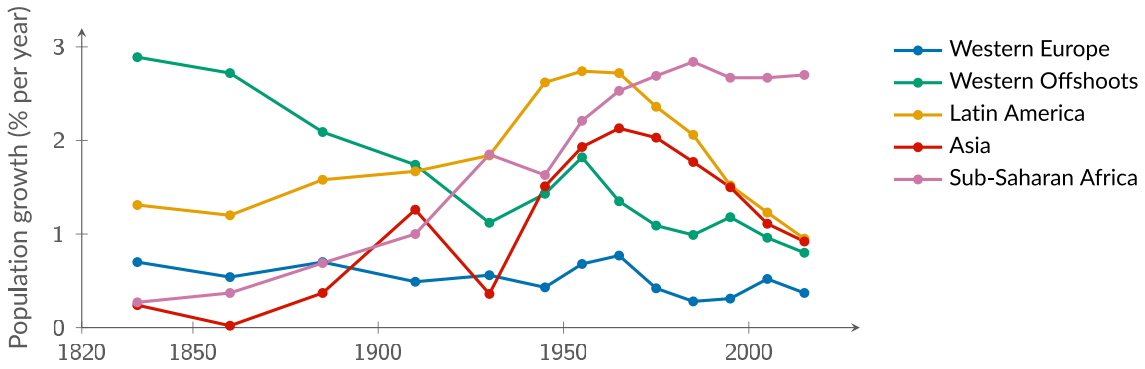


Figure 2: Population growth by region.

Note: g_N: average annual population growth (% per year) between benchmark years, plotted at interval midpoints.

4.2.3. Fertility over time

Total fertility rate (TFR) is lifetime births per woman at current age-specific fertility rates. Fertility declines at different times; several countries fall below replacement.

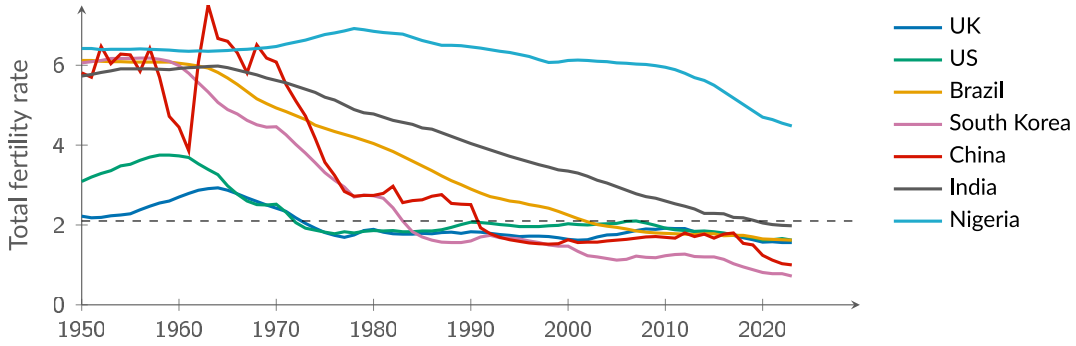


Figure 3: Fertility over time.

Note: TFR_t: total fertility rate (births per woman). Dashed line: replacement benchmark, about 2.1 births per woman.

4.2.4. Fertility and income across countries

In 2022, richer countries generally have lower fertility. Israel lies above the high-income cluster.

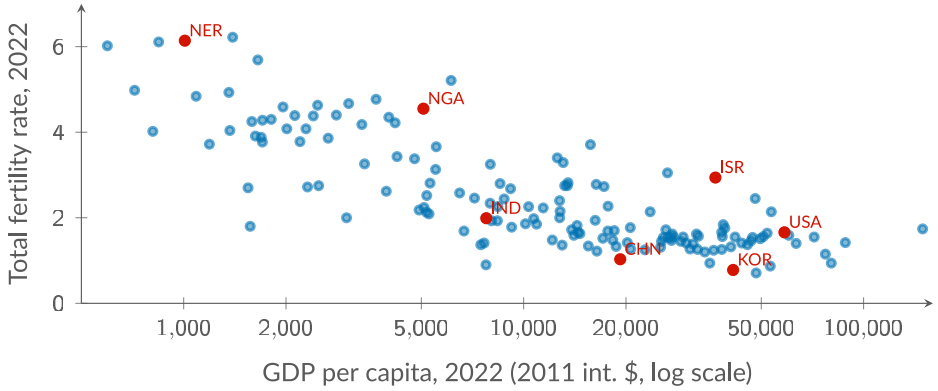


Figure 4: Fertility and income, 2022.

Note: TFR: births per woman; $y = \frac{Y}{N}$: income per capita in 2011 international dollars (log scale). Countries with population above one million.

4.3. Malthus to Solow: simulated transition

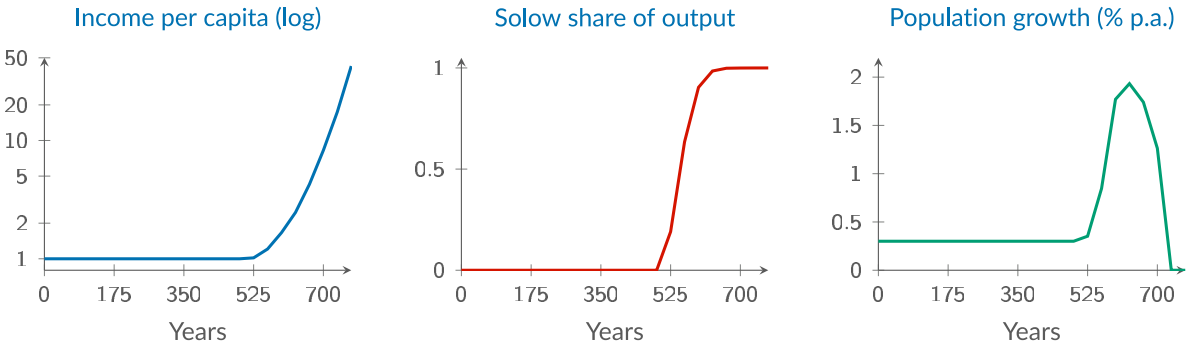


Figure 5: Simulated transition from Malthus to Solow.

Note: Left: income per capita $y = \frac{Y}{N}$ (model units, log scale). Middle: Solow output share $\frac{Y_S}{Y_M+Y_S}$. Right: population growth g_N (% per year). Time is in years; one model period is 35 years. Parameters: $(\alpha_1, \alpha_2, \theta) = (0.1, 0.6, 0.4)$; gross TFP growth factors per period: $(\gamma_M, \gamma_S) = (1.032, 1.518)$. Lecture 5, slide 12: simplified simulation using Hansen–Prescott technology parameters and population function (Hansen & Prescott, 2002), with a fixed saving rate in place of the OLG saving problem.

1. Initially, only the **Malthus sector** operates. Income per capita is flat; productivity gains raise population.
2. Rising Solow productivity crosses the **entry threshold**. The Solow sector's output share increases and income per capita starts to grow.
3. Population growth first rises, then falls. The decline comes from the **calibrated population function**; fertility choices are not derived here.

Lecture addition: Why an unused technology can improve

Solow-sector productivity grows even before the sector operates. The lecture interpreted this as accumulated knowledge from other activities: precise metalworking and other complementary skills eventually make industrial production viable. The steam engine illustrates how an invention can depend on many earlier improvements. Productivity growth is still exogenous in the model.

Recording: 2026-10-12, 00:54:30 - 00:56:12

Lecture addition: Trade and the return to education

A student asked how trade affects the transition. Galor and Mountford (Galor & Mountford, 2008) study specialization and demographic change:

- An early industrializer specializes in activities intensive in human capital and imports food. Higher returns to education favor investment per child.
- A later industrializer specializes in agriculture. Lower returns to education favor higher fertility and can delay its transition.

Trade can therefore widen differences in income per person and population growth through the quantity-quality choice.

Recording: 2026-10-12, 00:59:39 - 01:01:58

Lecture addition: Food and structural change

Hansen and Prescott treat the two sectors' outputs as perfect substitutes. If food must instead come from agriculture, a growing population may draw workers back into the land-dependent sector. Feeding the population then becomes a constraint on the expansion of industry.

Recording: 2026-10-12, 01:02:24 - 01:03:03

4.4. Becker, Murphy and Tamura (1990)

Human Capital, Fertility, and Economic Growth (Becker et al., 1990). Lecture 5, slides 14 to 28 use a simplified version without physical capital.

Parents choose the number of children and education per child: the *quantity-quality choice*. The model can have two stable regimes: **high fertility and stagnation**, or **lower fertility and sustained growth**. Escaping stagnation requires a sufficiently large shock.

4.4.1. Demographics, state and choices

People live for two periods: childhood, then adulthood. Adults work and consume, and spend time raising and teaching the next generation.

- State: acquired human capital $H \geq 0$. Every adult also has raw-labor productivity $H_0 > 0$; total productivity is $H_0 + H$.

- Choices: consumption $c > 0$, number of children $n > 0$ (continuous), and teaching time per child $h \geq 0$.
- Next state: each child inherits the same human capital H'.
- Adult time endowment is one. Each child requires fixed childcare time $v > 0$ and a goods cost $f \geq 0$.

This version has no assets or aggregate shocks. H_0 is fixed raw-labor productivity, distinct from the initial value of H.

4.4.2. Dynastic preferences

Parents value their own consumption and their children's utility:

$$\begin{aligned} u(c) &= \frac{c^\sigma}{\sigma}, \quad 0 < \sigma < 1, \\ a(n) &= \alpha n^{-\varepsilon}, \quad \alpha > 0, \quad 0 < \varepsilon < 1, \\ V(H) &= u(c) + a(n) n V(H') \\ &= \frac{c^\sigma}{\sigma} + \alpha n^{1-\varepsilon} V(H'). \end{aligned} \tag{4.1}$$

- Altruism **per child**, $a(n)$, falls with fertility; total weight on children, $\alpha n^{1-\varepsilon}$, rises.
- $\varepsilon = -\frac{d \log a(n)}{d \log n}$ measures how altruism per child falls with fertility. Relative risk aversion is $1 - \sigma$.

Assume finite dynastic utility. The continuation weight varies with n.

> **Lecture addition: The continuation value belongs to the child**
>
> $V(H')$ is the child's adult utility, including that child's concern for the next generation. Parents care about their children's actual welfare. If H' differs from parental human capital, the child can choose different fertility and education. Identical people within a generation need not make identical choices across generations.
>
> Recording: 2026-10-12, 01:06:07 - 01:08:09; 01:11:08 - 01:11:59

4.4.3. Resources and human-capital production

Childcare and teaching leave $1 - n(v + h)$ units of working time:

$$\begin{aligned} c + fn &= [1 - n(v + h)](H_0 + H), \\ H' &= Ah(H_0 + H), \quad A > 0, \\ n(v + h) &\leq 1. \end{aligned} \tag{4.2}$$

A is teaching productivity. Higher parental human capital raises both the opportunity cost of time and the amount a parent can teach.

Eliminate h to write the problem in (n, H'):

$$
\begin{aligned}
h &= \frac{H'}{A(H_0 + H)}, \\
c(H, n, H') &= (1 - nv)(H_0 + H) - \frac{nH'}{A} - fn.
\end{aligned}
\tag{4.3}
$$

The Bellman equation is

$$
V(H) = \max_{n>0, H' \geq 0} \left\{ \frac{c(H, n, H')^{\sigma}}{\sigma} + \alpha n^{1-\varepsilon} V(H') \right\}, \quad c(H, n, H') > 0. \tag{4.4}
$$

More children raise the total cost of providing a given education per child.

4.4.4. Optimal education: corner condition and Euler equation

At a differentiable optimum, the marginal benefit cannot exceed the marginal cost at zero investment; they are equal at positive investment:

$$
\begin{aligned}
\alpha n^{1-\varepsilon} V'(H') &\leq \frac{nc^{\sigma-1}}{A}, \\
H' \left[\frac{nc^{\sigma-1}}{A} - \alpha n^{1-\varepsilon} V'(H') \right] &= 0.
\end{aligned}
\tag{4.5}
$$

The envelope condition is

$$
V'(H) = c^{\sigma-1}(1 - nv). \tag{4.6}
$$

Holding H' fixed keeps teaching cost $n\frac{H'}{A}$ independent of H. The envelope coefficient $1 - nv$ therefore differs from working time $1 - n(v + h)$.

Let c' and n' denote the child's adult choices. Substitute the child's envelope condition into the education FOC:

$$
\begin{aligned}
\alpha n^{1-\varepsilon} (c')^{\sigma-1} (1 - n'v) &\leq \frac{nc^{\sigma-1}}{A}, \\
\underbrace{\frac{n^{\varepsilon}}{\alpha} \left(\frac{c'}{c} \right)^{1-\sigma}}_{\text{consumption MRS}} &\geq \underbrace{A(1 - n'v)}_{\text{education return}}.
\end{aligned}
\tag{4.7}
$$

Equality holds if $h > 0$. High fertility raises $\frac{n^{\varepsilon}}{\alpha}$ and discourages investment in each child's future consumption.

Lecture addition: Why education can be zero

Children receive H_0 without teaching. When acquired human capital is low, a small addition to H' gives a small proportional gain in their future productivity, while teaching immediately reduces parental working time. Because parents also pay the fixed food cost fn, the percentage loss in their consumption can exceed the percentage loss in earnings.

When acquired skills account for most of children's future earnings, education has a much larger effect on their welfare. This is the intuition behind the two investment regimes.

Recording: 2026-10-12, 01:14:03 - 01:16:24

4.4.5. Optimal fertility

For an interior fertility choice, differentiate the Bellman equation with respect to n:

$$\underbrace{(1-\varepsilon)\alpha n^{-\varepsilon}V(H')}_{\text{benefit of another child}} = \underbrace{c^{\sigma-1}[(v+h)(H_0+H)+f]}_{\text{goods and time cost}}. \tag{4.8}$$

Higher H makes childcare and teaching time more expensive. The substitution effect favors fewer children and more education per child. Higher income also raises demand for children, so the net fertility response depends on both effects.

Lecture addition: Positive fertility and the quantity-quality choice

At a fixed continuation value $V(H') > 0$, the marginal benefit of children diverges as fertility approaches zero:

$$(1-\varepsilon)\alpha n^{-\varepsilon}V(H') \to \infty \quad \text{as} \quad n \to 0.$$

This supports positive fertility under the model's assumptions, even when teaching is zero.

Planning more education per child raises the time cost of every additional child through $(v+h)(H_0+H)$. Education choices make children more expensive and can reduce fertility.

Recording: 2026-10-12, 01:16:24 - 01:18:34

4.4.6. Malthusian steady state

Every generation chooses zero education:

$$\begin{aligned} H = H' = h = 0, \\ c_M = H_0 - n_M(vH_0 + f) > 0. \end{aligned} \tag{4.9}$$

With constant consumption and fertility,

$$\begin{aligned} V(0) &= \frac{c_M^\sigma}{\sigma} + \alpha n_M^{1-\varepsilon} V(0), \\ &= \frac{c_M^\sigma}{\sigma\left[1 - \alpha n_M^{1-\varepsilon}\right]}. \end{aligned} \tag{4.10}$$

Finite utility requires $\alpha n_M^{1-\varepsilon} < 1$. Fertility n_M satisfies the FOC below; substituting $V(0)$ gives its second form:

$$(1-\varepsilon)\alpha n_M^{-\varepsilon} V(0) = c_M^{\sigma-1}(vH_0 + f). \tag{4.11}$$

$$\frac{(1-\varepsilon)\alpha n_M^{-\varepsilon} c_M}{\sigma\left[1 - \alpha n_M^{1-\varepsilon}\right]} = vH_0 + f. \tag{4.12}$$

For a **strict no-investment corner**, the Euler condition becomes

$$\frac{n_M^\varepsilon}{\alpha} > A(1 - n_M v). \tag{4.13}$$

- High fertility raises the left side and lowers the right side. Suitable v, f and preference parameters support the no-investment corner.
- The strict inequality persists after small increases in H. Parents still choose $H' = 0$, returning to stagnation in one generation.
- Consumption per adult stays constant. Under the model's one-adult household convention, $N_{t+1} = n_M N_t$; population grows if $n_M > 1$.

4.4.7. Balanced growth path

For sufficiently large H, education can sustain growth. Along a growing path, $\frac{H_0}{H} \to 0$ and $\frac{f}{c} \to 0$. Set $H_0 = f = 0$ in the limiting problem:

$$\begin{aligned} c &= [1 - n(v+h)]H, \\ H' &= AhH. \end{aligned} \tag{4.14}$$

On the **balanced growth path** (BGP), n and h are constant. Define the gross growth factor $1 + g$:

$$1 + g = \frac{c'}{c} = \frac{H'}{H} = Ah. \tag{4.15}$$

Homogeneity implies $V(H) = CH^\sigma$, with $C > 0$, so

$$\frac{V'(H')H'}{V(H')} = \sigma. \tag{4.16}$$

The interior FOCs for teaching and fertility are

$$\begin{aligned} A\alpha n^{-\varepsilon} V'(H') &= c^{\sigma-1}, \\ (1-\varepsilon)\alpha n^{-\varepsilon} V(H') &= c^{\sigma-1}(v+h)H. \end{aligned} \tag{4.17}$$

Divide the first by the second, then multiply by H':

$$\begin{aligned} \frac{1}{1-\varepsilon}\frac{V'(H')H'}{V(H')} &= \frac{H'}{AH(v+h)}, \\ \frac{\sigma}{1-\varepsilon} &= \frac{h}{v+h}, \\ (1-\varepsilon-\sigma)h &= \sigma v. \end{aligned} \tag{4.18}$$

$$\begin{aligned} h &= \frac{\sigma v}{1-\varepsilon-\sigma}, \\ 1+g &= \frac{A\sigma v}{1-\varepsilon-\sigma}. \end{aligned} \tag{4.19}$$

Positive teaching requires $1-\varepsilon-\sigma > 0$; positive growth requires $1+g > 1$. These conditions describe the limiting growth regime. An economy starting near $H = 0$ can still remain in stagnation.

Fertility on the BGP satisfies the Euler equation

$$\begin{aligned} \frac{n^{\varepsilon}}{\alpha}(1+g)^{1-\sigma} &= A(1-nv), \\ n^{\varepsilon} &= \alpha A(1-nv)(1+g)^{\sigma-1}. \end{aligned} \tag{4.20}$$

The candidate must also satisfy

$$\begin{aligned} n(v+h) &< 1, \\ \alpha n^{1-\varepsilon}(1+g)^{\sigma} &< 1. \end{aligned} \tag{4.21}$$

The first condition leaves time for production and positive consumption. The second ensures finite dynastic utility as consumption grows:

$$V(H) = \frac{[1-n(v+h)]^{\sigma}H^{\sigma}}{\sigma\left[1-\alpha n^{1-\varepsilon}(1+g)^{\sigma}\right]}. \tag{4.22}$$

- On an interior feasible BGP, higher fixed childcare time v raises teaching time per child and the growth rate.
- α affects BGP fertility; it drops out of the growth formula.
- Growth is **endogenous**: parents choose teaching while A stays constant. With $H_0 > 0$ or $f > 0$, the BGP formulas hold asymptotically.

Lecture addition: Mass schooling and fertility

The lecture linked falling fertility in today's high-income countries to the expansion of mass schooling in the late nineteenth century. Education increased the investment families made in each child, motivating the quantity-quality mechanism in these growth models.

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4.4.8. Two regimes and the transition threshold

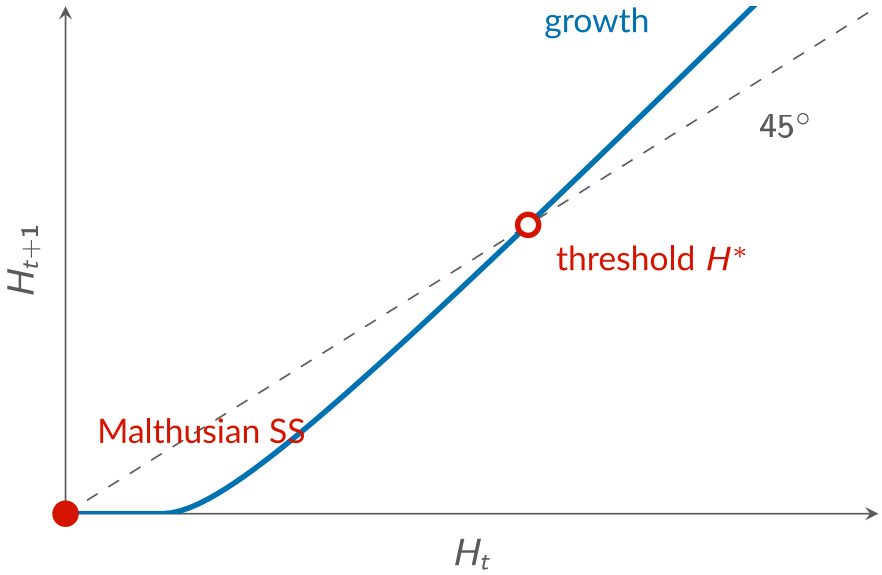


Figure 6: Human capital and the two regimes.

Note: H_t and H_{t+1}: acquired human capital per adult in successive generations (model units). Dashed line: $H_{t+1} = H_t$. H^*: threshold separating stagnation and growth in the schematic. Lecture 5, slide 27. The curve is a schematic, not a calibrated policy function.

- Near zero, $h = 0$ and $H_{t+1} = 0$.
- Below the threshold, positive education can still be insufficient to replace parental human capital: $H_{t+1} < H_t$.
- Above the threshold, $H_{t+1} > H_t$; the economy approaches the BGP. Fertility is lower and teaching per child is higher in the growth regime.
- A large enough shock to human capital or technology can change the eventual regime. A small temporary increase in H may leave it unchanged.

On the BGP, **growth rates and time allocation** stay constant while human capital keeps rising. The smooth threshold crossing is schematic: a nonconcave problem can have a policy jump or a kink in V. FOCs alone do not establish the global optimum.

4.4.9. Assessment

- Cheap parental time favors many children. Costly time and productive teaching favor more investment per child.
- The *Malthusian* regime has high fertility and stagnant income per adult. Land is absent, and there is no exogenous productivity trend whose gains are absorbed by population growth.
- A large shock can start sustained growth. Its historical origin is outside the model.
- Stagnation can be optimal for a dynasty under these preferences and constraints. Pareto efficiency across generations requires a specified welfare criterion; it does not follow from household optimality alone.
- The dynastic-altruism framework remains widely used to model fertility in macroeconomics.

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